TECHNICAL INTERVIEW PORTFOLIO

AI Hardware Architecture & Simulation
EDA Tool Development & Verification

AI Memory Systems / Software Simulation / EDA Tool Development

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AI Hardware Architecture / GPU–HBM/HBF Co-design

Education and Technical Background

01 / BACKGROUND

Education

National Taiwan University
M.S. in Data Science
Sep. 2025 – Expected Jun. 2027

National Chengchi University
B.S. in Computer Science
Double Major in Electrophysics
2021.09 – 2025.06

Research and Engineering

AI Hardware Systems
GPU / HBM / HBF Memory Hierarchy
Workload-Aware Performance Modeling
HW/SW Co-design Exploration

EDA Software Development
Logic Synthesis / Technology Mapping
C++ Tool Integration / Verification

Professional Experience

02 / INDUSTRY EXPERIENCE

NeoPower

Jun. 2023 – Mar. 2025

Contract Software Engineer

Custom Simulator: Built a simulator to reproduce operational workflows and error scenarios.

Software Validation: Performed end-to-end testing, regression testing, and debugging.

Customer-Side Validation: A/B testing, post-deployment checks, and log/data analysis.

Smart Mobility Subsidiary (Inventec Group)

Apr. 2025 – Jul. 2025

Contract Software Engineer

Contributed to software development, system integration, testing, and debugging.

Projects and Research Focus

03 / PORTFOLIO ROADMAP

AI hardware architecture first, followed by EDA software development and validation.

01 AI Hardware Systems

Workload-Aware HBM/HBF Memory Architecture Exploration

MoE Inference / Hardware Modeling / Performance–Power–Endurance Trade-offs

02 EDA Tool Development

Multi-Output Standard-Cell Technology Mapping

C++ / Boolean Matching / Optimization / Correctness Validation

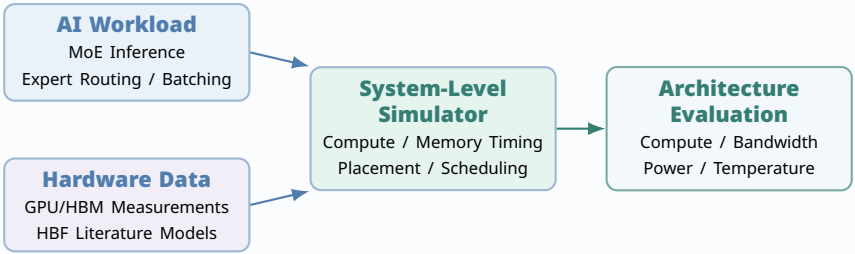
Shared skills: Simulation, Software Development, Quantitative Evaluation, and Testing.

04 / AI SYSTEMS – PROJECT INTRODUCTION

Workload-Aware GPU Memory Architecture

Hardware/Software Co-design / System-Level Simulation

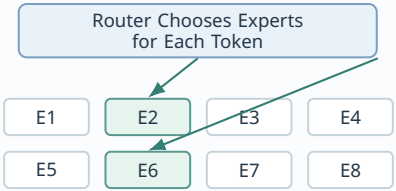
Co-design GPU memory architectures for the compute and memory needs of targeted AI models.



Simulation uses available hardware specifications and measurements; HBF uses published models and design parameters.

MoE Inference: Memory Bottleneck

MoE reduces computation per token, but large expert weights must still be stored and transferred as needed.



Only selected experts are activated;
all expert weights still need storage.

Compute Efficiency

Each token activates only a subset of experts, reducing computation.

Memory Capacity

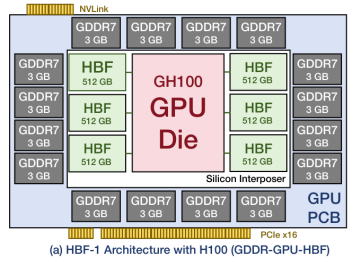
The full set of expert weights may exceed HBM capacity.

Transfer Bottleneck

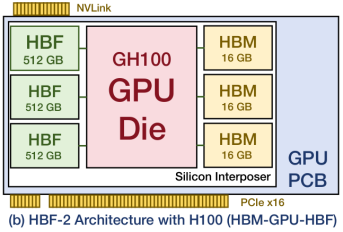
Loading experts from host DRAM / SSD may stall GPU execution.

Hybrid HBM/HBF: Candidate Memory Architecture

Trade some HBM stack sites for higher HBF capacity and explore package-local memory trade-offs.



(a) HBF-1(2028)



(b) HBF-2(2030)

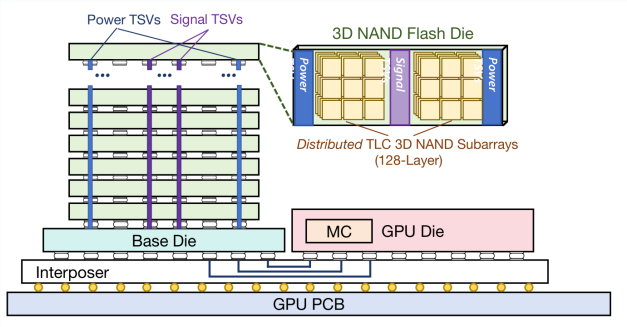
Modeled HBF-2 Setup
3 HBM + 3 HBF; 48 GB HBM + 1.5 TB HBF.

Capacity Benefit
Larger on-package memory capacity.

Architecture Trade-off
Reduced HBM capacity and bandwidth.

HBF Physical Organization: 2.5D/3D Packaging

HBF integrates stacked NAND, TSVs, a base die, and an interposer in an advanced package.



Software Simulation: Architecture Exploration

Use software simulation to compare memory configurations and MoE workloads before hardware implementation.

Workload Inputs
MoE model, expert selection, request arrivals, context length, and batch size.

Hardware Parameters
GPU compute, stack count, capacity, latency, bandwidth, and power.

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System-Level Simulator
Compute / Transfer Timing

Data Placement / Migration

Scheduling / Prefetch

Thermal / Endurance Models

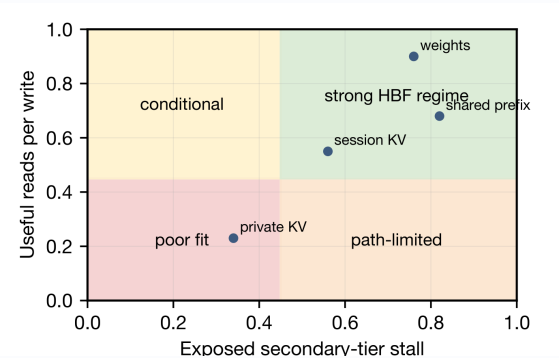
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Evaluation Metrics
Throughput, TTFT/TPOT, SLO goodput, bandwidth, temperature, and lifetime.

Software Policies
Weight placement, batching, and HBM watermark parameter sweeps.

HW/SW Co-design: Data Placement and Constraints

Capacity alone is not enough: data reuse, transfer stalls, power, and write endurance also matter.



HBM: Latency-Sensitive Data

Keep active experts and frequently updated KV cache in HBM.

HBF: Read-Mostly Data

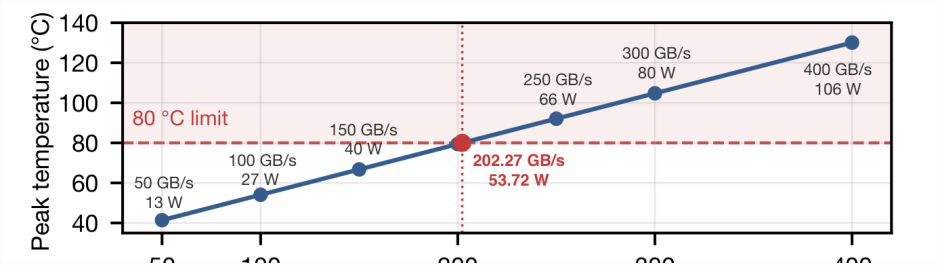
Evaluate expert weights and reusable shared prefixes for HBF.

Policies and Reliability

Explore prefetching, scheduling, HBM watermarks, thermal caps, and P/E wear.

Thermal-constrained HBF Bandwidth

Sustained bandwidth is limited by thermal constraints, not just peak interface bandwidth.



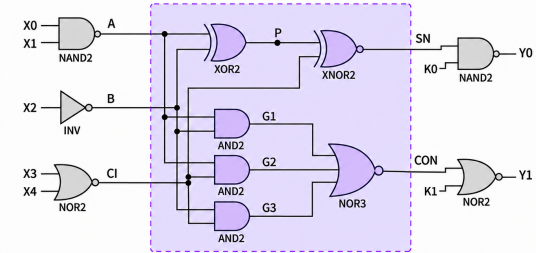
Literature Model

In the paper model, a 16-Hi TLC HBF stack sustains about 202 GB/s near 80 °C.

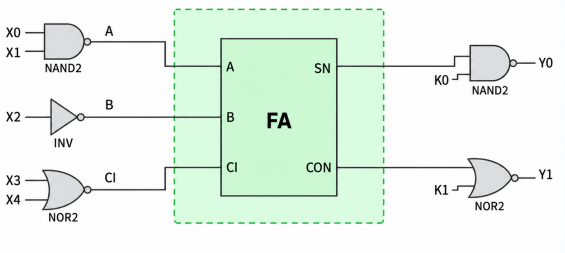
Technology Mapping: Why Multi-Output Cells?

Logic synthesis maps functions to standard cells; multi-output cells may share logic.

Single-Output Mapping



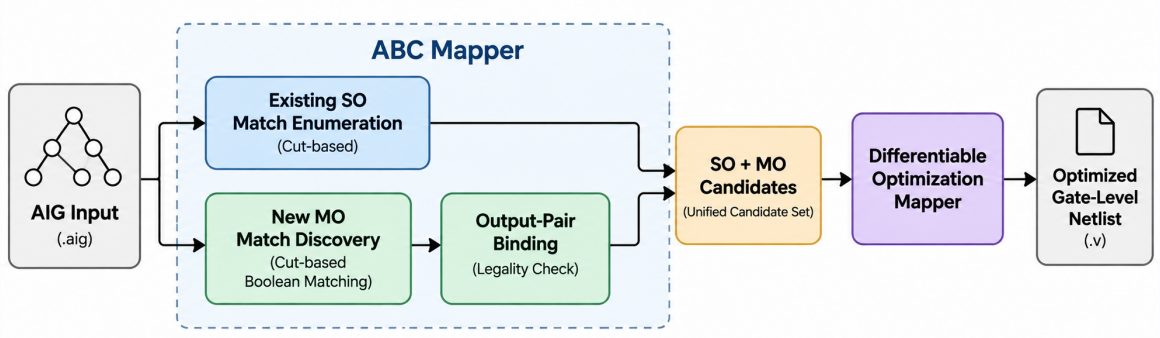
Multi-Output Mapping



Potential benefit: Shared logic may reduce both area and delay.

ABC Mapper: Multi-Output Mapping Extension

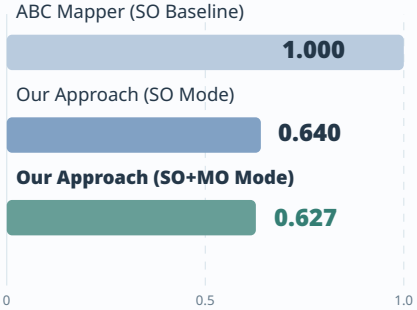
Added multi-output candidate discovery and output-pair binding to ABC.



Experimental Results: Area-Delay Product

13 / EDA PROJECT – EVALUATION 30 benchmark cases | 7nm ASAP7 | GMean normalized ADP

Normalized ADP (Area–Delay Product)



37.3%

ADP reduction vs. ABC baseline

2.0%

Further improvement vs. SO mode

The 30-case results are from an earlier bridge-based flow; ABC-native MO discovery was developed later.

Software Engineering: Verification and Evaluation

Preserve logic correctness and benchmark reproducibility when optimizing tools.

Correctness

Regression Testing
Start with small smoke tests after each change.

Boolean Equivalence
Use combinational equivalence checking (CEC) to compare the original circuit and mapped netlist.

Backward Compatibility
Preserve the existing single-output mapping flow.

Reproducibility

Quantitative Comparison
Use fixed benchmarks and baselines for controlled comparisons.

Hardware-Aware Metrics
Use cell-library area/delay and normalized ADP.

Resource Discipline
Bound the search space; monitor runtime and peak memory.

Thank You

Q&A